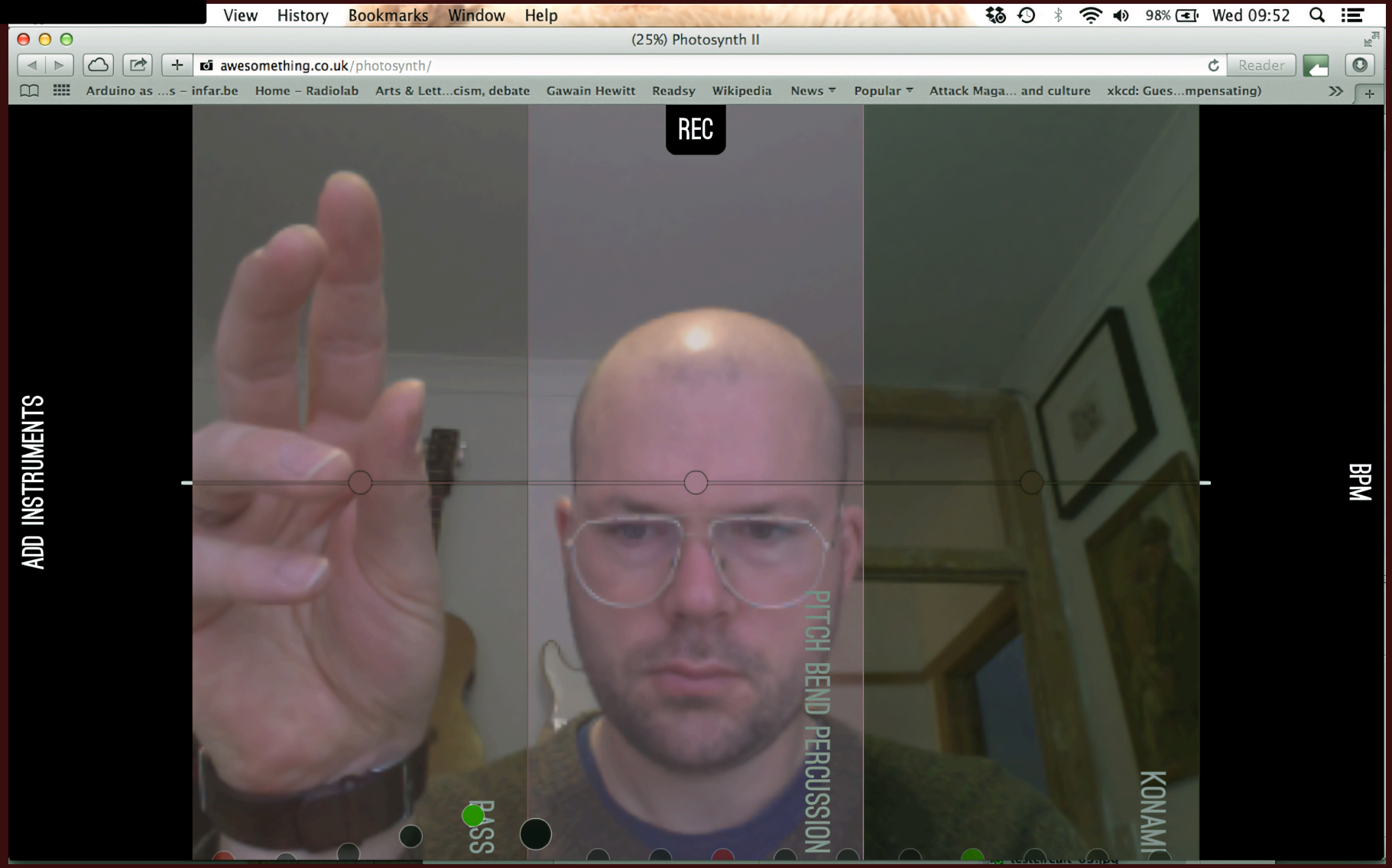
PHOTOSYNTH

SMILE POWERED MUSIC

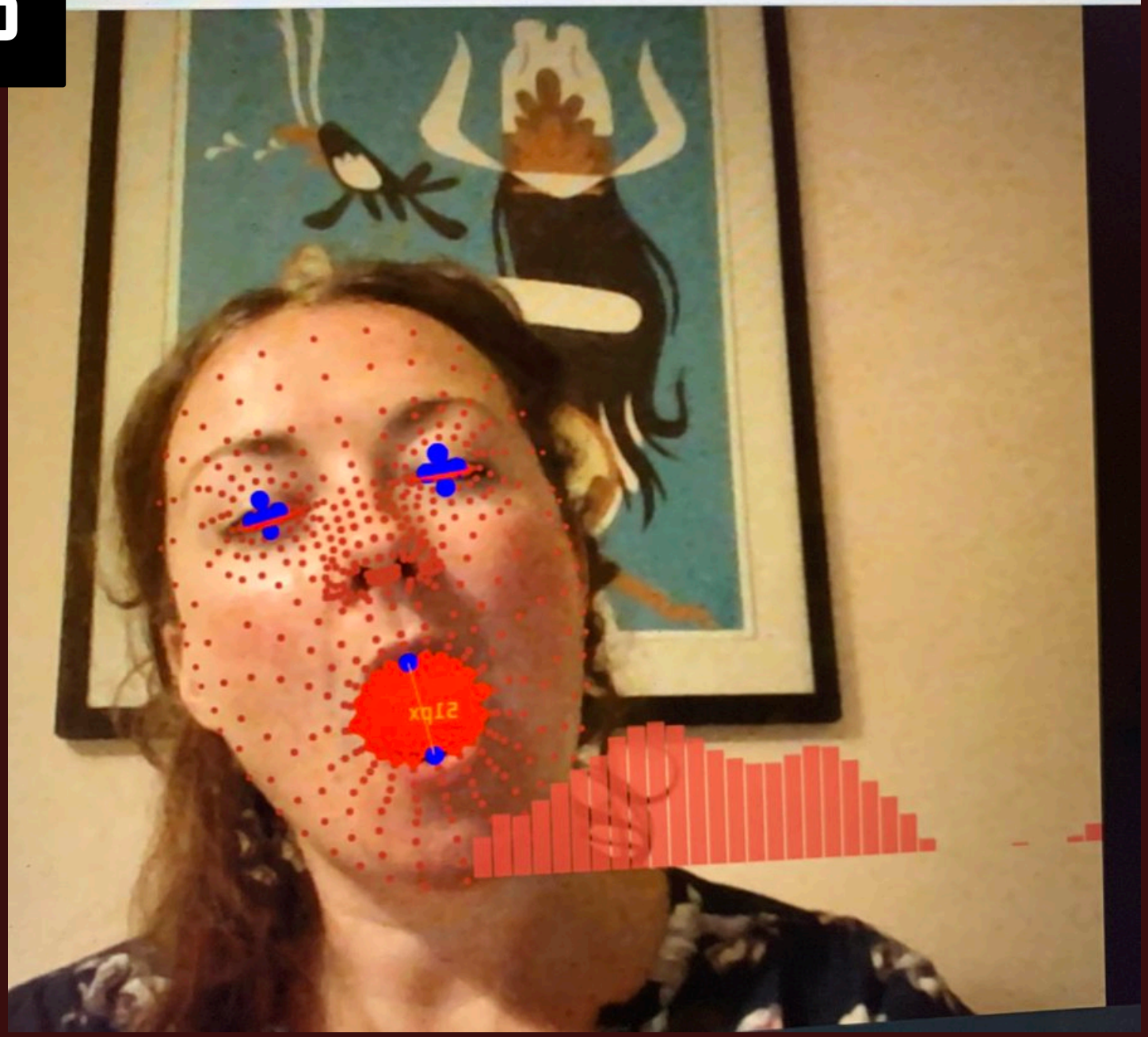
What if rather than learning how to play a musical instrument, the musical instrument learns how to play you?

2011



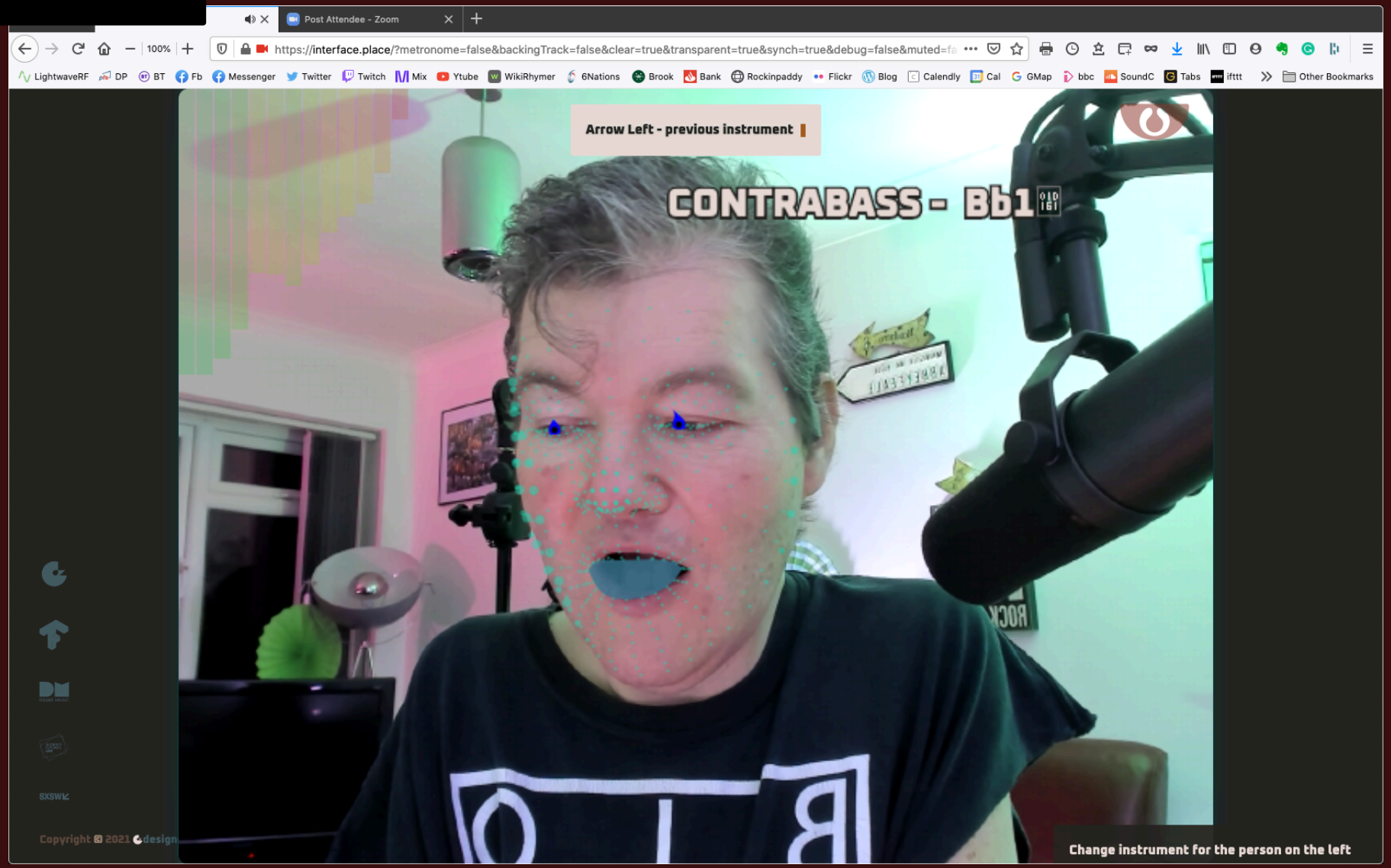
Gawain Hewitt plays the PhotoSYNTH II

2020



Becky Morris Knight plays with a PhotoSYNTH III Beta during Lockdown

2021



John Kelly demonstrates the PhotoSYNTH III at #DMLab

2021

PhotoSYNTH III Beta being demonstrated by Louis from Drake Music

2022

WAC 2022 showcase of PhotoSYNTH III

2022

Collaborative MIDI Music over Zoom using PhotoSYNTH III and OP-1

2022

PhotoSYNTH at the MIDI Innovation Awards Finals

Constant testing with real people ensured that it evolved in an accessible and ergonomic way

Everyone can make music friends.

With the broadest demographic covering every diversity and ability...

the PhotoSYNTH levels the playing field for any aspiring creatives!

12. **July 2024,** started working with Future Everything
11. **June 2024,** *PhotoSYNTH4D*, the holographic version, was publicly demonstrated for the first time at the Electromagnetic Fields festival '24.
10. **November 2023,** demonstrated live at a #DMLab event and received a grant for further development from Drake Music.
9. **September 2022,** an early prototype was publicly presented as an installation at the Web Audio Conference '22.
8. **June 2022,** *PhotoSYNTH3#interFACE* was featured at the MIDI Innovation Award
7. **March 2022,** *PhotoSYNTH3#interFACE* was presented at Music Maker Festival
6. **March 2021,** presented a talk about the *PhotoSYNTH3 #interFACE* at SXSW21 music technology conference titled, *'Give a man a record and they dance for day, give that man a synthesizer and they dance for a lifetime'*
5. **February 2021,** represented the UK at Audience of the Future
4. **January 2021,** *PhotoSYNTH3#interFACE* receives Creative Industries Council's Createch “One to Watch” winner 2021
3. **December 2020,** *PhotoSYNTH3#interFACE* was first presented at the Xmas DMLab event. (Alpha version being demoed by Tim Yates from DMLab)
2. **October 2014,** *PhotoSYNTH II* was presented at the DMHackDay accessible technology contest
1. **November 2011,** *PhotoSYNTH* was presented at Music Hack Day 2011

2024

March 2024

1st International Workshop on the Musical Metaverse

IEEE showcase of PhotoSYNTH 4D

July 2024

Electromagnetic Fields Festival demo of HOLOGRAPHIC PhotoSYNTH 4D

July 2024

Electromagnetic Fields Festival group MIDI demo of PhotoSYNTH 4D

September 2024

Accessible Musical Instrument Collection PhotoSYNTH 4D Holographic

FUTURE EVERYTHING

CULTURAL ACCELERATOR

Funding kickstarted the development of **Quartet** variant.

NEW 4 PLAYER MODE UNLOCKED!

NEW 4 PLAYER MODE UNLOCKED!

NEW 4 PLAYER MODE UNLOCKED!

NEW 4 PLAYER MODE UNLOCKED!

NEW 4 PLAYER MODE UNLOCKED!

November 2024

Beyond '24 demonstration of 4-Player PhotoSYNTH IV

January 2025

MIT demonstration of PhotoSYNTH IV in Quartet mode

SCIENTIFIC PAPERS BASED ON THIS WORK :

- PhotoSYNTH3 - A novel interFACE (2022)
- Instant Musical Superpowers (preprint) 2023)
- Instant musical superpowers by augmenting intrinsic movements, expressions and emotions (2023)
- Musical Avatars (2024)

Musical avatars

1st Zenon Olenski
Member
designerzen (of designerzen.com)
London / Manchester, UK
zenon@designerzen.com

***Abstract*—In this paper, we introduce and describe *expressive ways to control machines using parts of the body*, predominantly through gauging intent from facial expressions, reflexive locomotion, and the orientation and motions of the head and body—without requiring any physical touch or interaction.**

We then present a fully working implementation in the form of a WebXR based, multiplayer face-controllable online musical instrument that demonstrates how this simplified form of input can be highly expressive yet still retain fine control, subtlety, and finesse.

We also suggest researched methods for more inclusive human-machine interactions, where accessibility, expression and fidelity are the key motivators.

***Index Terms*—musical avatar, xr, vr, metamusic, metaverse, audio, music, performance recording, performance capture**

I. Introduction

Through repetition our brains associate actions with outcomes, allowing us to learn new skills [1]. This paper and the related research projects postulate that if a desired skill can be mapped onto natural movements already employed by the body and familiar in daily life, then the time to learn this new skill will be vastly reduced and repeatable with more precision and fewer errors, allowing for greater expression, range and fidelity.

To research this theory we have created an expressive, online, extended-reality musical instrument, playable entirely with the face, that requires no specialized hardware or equipment, training or previous skills.

By observing a person's own natural movements using motion-tracking in real-time, the software creates a personalized musical avatar that reacts to every movement and expression. The full three-dimensional expression, including any personal style, flair and precision, is saved as part of the performance where it can be recalled in full definition.

Collaboration is encouraged and built in at a foundational level for use cooperatively in both the real world through augmentation, and within the metaverse as new avatar abilities.

We demonstrate how to operate an interface by mapping simple, obvious, movements onto a complex control system in an intuitive and ergonomic way, and how, by observing subtle emotional cues and facial expressions, we can comprehend intention and mood using a traditional understanding of human psychology and use this to modify the data and musical output accordingly.

Drake Music Charity #DMLab

This accessible instrument aims to grant new musical abilities to those unable or unwilling to use existing instruments, by trying to find simpler and more frictionless ways of controlling technology through augmented integration.

Expressions, movements and emotions can be mapped to control and manipulate data, software and hardware.

We are continuously running analytical studies and experiments at the Accessible Musical Instrument Collection (AMIC) where we iterate and refine the control mechanism based on the feedback and metrics from disabled users. This ensures that the software works for the highest percentile of humans with the broadest range of diversity. The approach outlined in this paper represents the ongoing conclusions from this research and describes a refined, considered approach to input that balances ergonomics with ease of use without compromising on fidelity.

By training machine learning models on human forms, we can use them to observe video data and understand basic intentions and emotions in near real-time, which, over time, can learn and improve as it gets more familiar with, and better at understanding your intentions and how you choose to express them.

Using the face as a control mechanism for music feels familiar as it is intrinsically linked to singing—so for this pilot project we convert smiles into musical notes, movements into melodies and facial expressions into audio filters and controls.

II. Operational Requirements

This virtual interface requires no specialist hardware and can run on most modern computers and mobile devices with dedicated GPUs.

- From 1 to 4 people (per device)
- Video Camera (webcam / phone camera)
- Web browser that supports GPU operations
- *Screen / Projector / Holographic display*
- Modern computer or mobile device
- *Input device / touch screen* (optional)

Any modern laptop or phone that satisfies the above requirements should work. *A mouse, keyboard or touch screen interface is required to initiate play but only a face and mouth is necessary once the application has begun.*

III. Demonstration

An accessible musical synthesizer, completely controllable with the face, has been made publicly available as a permanent

BREAKING NEWS!

THIS SUMMER, WE WIL BE PRESENTING THE PHOTOSYNTH AT THE ROYAL SOCIETY SUMMER EXHIBITION.

2025 Summer Exhibition @ Royal Society

IMPORTANT!

LOOKING FOR FUNDS FOR STAFF, VOLUNTEERS AND SUPPORT

July 1st-7th 2025

Royal Society Summer Exhibition 2025